

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026–2027

Tutorial Sheet–3

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

(1) What will be the unit digit of the squares of the following numbers?

(i) 81 (ii) 272 (iii) 799 (iv) 3853 (v) 1234 (vi) 26387

(2) The squares of which of the following would be odd numbers?

(i) 431 (ii) 2826 (iii) 7779 (iv) 82004

(3) Observe the following pattern and find the missing digits.

$$\begin{aligned}
 11^2 &= 121 \\
 101^2 &= 10201 \\
 1001^2 &= 1002001 \\
 100001^2 &= 1\ldots\ldots 2\ldots\ldots 1 \\
 10000001^2 &= \ldots\ldots\ldots
 \end{aligned}$$

(4) Using the given pattern, find the missing numbers.

$$\begin{aligned}
 1^2 + 2^2 + 2^2 &= 3^2 \\
 3^2 + 4^2 + 12^2 &= 13^2 \\
 4^2 + 5^2 + _^2 &= 21^2 \\
 5^2 + _^2 + 30^2 &= 31^2 \\
 6^2 + 7^2 + _^2 &= _^2
 \end{aligned}$$

(5) How many numbers lie between squares of the following numbers?

(i) 12 and 13 (ii) 25 and 26 (iii) 99 and 100

(6) Verify that the following equality holds for any integer $m \geq 0$:

$$(2m)^2 + (m^2 - 1)^2 = (m^2 + 1)^2$$

In this case, the triplet $(2m, m^2 - 1, m^2 + 1)$ is known as *Pythagorean triplet*.

(7) Write a Pythagorean triplet whose smallest member is 8.

(8) Write a Pythagorean triplet whose one member is:

(i) 6 (ii) 14 (iii) 16 (iv) 18

(9) Find the smallest square number which is divisible by each of the numbers 6, 9 and 15.

(10) What could be the possible ‘one’s’ digits of the square root of each of the following numbers?

(i) 9801 (ii) 99856 (iii) 998001

(11) Without doing any calculation, find the numbers which are surely not perfect squares.

(i) 153 (ii) 257 (iii) 408 (iv) 441

(12) Find the square roots of the following numbers by the Prime Factorisation Method.

(i) 153 (ii) 1764 (iii) 4096 (iv) 7744

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- (13) For each of the following numbers, find the smallest whole number by which it should be multiplied so as to get a perfect square number. Also find the square root of the square number so obtained.
 (i) 252 (ii) 1008 (iii) 2028 (iv) 1458
- (14) Find the square root of each of the following numbers by Division method.
 (i) 2304 (ii) 4489 (iii) 3481 (iv) 5776
- (15) Find the least number which must be subtracted from each of the following numbers so as to get a perfect square. Also find the square root of the perfect square so obtained.
 (i) 402 (ii) 1989 (iii) 3250 (iv) 825
- (16) Find the least number which must be added to each of the following numbers so as to get a perfect square. Also find the square root of the perfect square so obtained.
 (i) 525 (ii) 1750 (iii) 252 (iv) 1825
- (17) Find the smallest number by which each of the following numbers must be multiplied to obtain a perfect cube.
 (i) 243 (ii) 256 (iii) 72 (iv) 675
- (18) You are told that 1,331 is a perfect cube. Can you guess without factorisation what is its cube root? Similarly, guess the cube roots of 4913, 12167, 32768.
- (19) Find the value of.
 (i) $(3^0 + 4^{-1}) \times 2^2$ (ii) $(2^{-1} \times 4^{-1}) \div 2^{-2}$ (iii) $(\frac{1}{2})^{-2} + (\frac{1}{3})^{-2} + (\frac{1}{4})^{-2}$ (iv) $(3^{-1} + 4^{-1} + 5^{-1})^0$
 (v) $\frac{8^{-1} + 5^3}{2^{-4}}$ (vi) $\left\{ \left(\frac{1}{3}\right)^{-1} - \left(\frac{1}{4}\right)^{-1} \right\}^{-1}$
- (20) Express the following numbers in usual form.
 (i) 3.02×10^{-6} (ii) 4.5×10^4 (iii) 3×10^{-8} (iv) 3.61492×10^6
- (21) Express the number appearing in the following statements in standard form.
 (a) 1 micron is equal to $\frac{1}{1000000}$ m
 (b) Charge of an electron is 0.000,000,000,000,000,16 coulomb.
 (c) Size of a bacteria is 0.0000005 m
 (d) Size of a plant cell is 0.00001275 m
 (e) Thickness of a thick paper is 0.07 mm
